LLMs & Fine-tuning

Train the Trainer: Gemma

Experts

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Build with AI

Fine-tuning

In this section

- What is LLM and how to fine-tune?
- The fine-tuning project lifecycle
- Fine-tuning strategies: Full vs Parameter-Efficient (PEFT)
- Scaling up: Distributed fine-tuning for massive models
- Practical application: Fine-tuning Gemma with Transformers and Keras
- Conclusion & Key Takeaways

LLM?

Currency of LLMs

What is Fine-tuning?

A type of post-training performed for **task adaptation** - the process of **updating the weights** of a pre-trained model to suit **a specific task or domain**.

This is achieved by training the model on **a smaller, more targeted dataset**.

Fine-tuning

What is it not?

Fine-tuning is not:

- **Training from Scratch:** Too expensive for most, requiring massive data and computation to train a model with random weights.
- **Prompt Engineering:** Guiding a pre-trained model's output via prompt instructions. It doesn't change model parameters; knowledge remains static but directed.

Fine-tuning

When should it be performed?

Key scenarios include:

- **Style & Persona Adaptation:** Adopting a specific voice or communication style.
- **Format Compliance:** Reliably generating structured outputs like JSON or XML.
- **Domain Specialization:** Improving accuracy and understanding of niche terminology.
- **Improving Reliability on Complex Tasks:** Correcting repeated failures to follow complex instructions that prompt engineering alone cannot fix.
- **Distillation:** Training a smaller, cheaper model to mimic the performance of a larger, more expensive one for a specific task to reduce cost and latency.

Fine-tuning

Is domain adaptation required?

If there is a need to familiarize the model with the specific vocabulary, concepts, and statistical patterns of a domain, **continued pre-training** before fine-tuning may be required.

This is a type of **domain adaptation.**

Fine-tuning

The fine-tuning project lifecycle

0 Strategy & Scoping

1 Data Preparation

2 Model & Strategy Selection

3 Experimentation & Evaluation

Formatting for fine-tuning

The way data is structured tells the model what it is expected to learn.

- **Instruction Tuning** (also known as **Supervised Fine-Tuning** or **SFT**) is the most common method. High-quality prompt/response pairs are provided to the model, which teaches it: "When you see this kind of prompt, produce this kind of output."
- **Preference Tuning:** To align a model with human preferences, the data is often formatted as a triplet: [prompt, chosen_response, rejected_response]. The teaches the model to produce outputs more like the "chosen" response and less like the "rejected" one.

Example: An Alien From Mars

Here we have an extract from the dataset we will be using later in this deck, for training Gemma 3 to talk like a hypothetical alien from Mars.

We can see that it is **formatted for instruction-tuning** - when the model sees certain player queries, it will be trained to respond with certain responses.

Dataset Viewer

Auto-converted to Parquet | API | Embed | Data Studio

Subset (2) martian · 25 rows

Split (1) train · 25 rows

Search this dataset

id int64	category string · classes	player string · lengths	alien string · lengths
1↔3 12%	Greetings … 20%	12↔16 12%	93↔105 20%
1	Greetings & Basic Questions	Hello there.	Gree-tongs, Terran. You'z a long way from da Blue-Sphere, yez?
2	Greetings & Basic Questions	Who are you?	I am Zorp. Juzt a humble rock-drinker from da Great Red-Dust. You... you z-mell like... wet…
3	Greetings & Basic Questions	What is this place?	Diz? Diz iz da Outpozt. A tiny piece of da Red-Dust on your... k'tak... very green world.
4	Greetings & Basic Questions	Are you friendly?	Friendly? k'tak Zorp iz not for eating you, if dat iz your meaning. You have shiny bitz?
5	Greetings & Basic Questions	Goodbye.	Vorp-vorp, Earth-kin. May your zky stay blue.
6	Quest-Related Dialogue	Do you need any help?	Yez! Yez! My Hydro-ziphon iz broken! Da crystal inside haz gone... k'tak... dark. Can…

Choosing a base model

Based on the objective and available resources, a base model is chosen as a starting point.

- Model size - small or large?
- Base pre-trained model or instruction-tuned variant?
- Desired performance characteristics?
- Licensing considerations?

Choosing a fine-tuning strategy

Based on the model chosen and the resources available, a tuning strategy is decided upon.

- **Full Fine-Tuning** for small models, or
- **Parameter-Efficient Fine-Tuning (PEFT)** methods for larger models to balance performance and efficiency

Full Fine-Tuning

Updates all weights and biases of the model by training on the new dataset.

Pros	• **Very Effective:** Potentially the highest performance and deepest task adaptation.
Cons	• **Computationally Expensive:** Requires significant GPU memory to store gradients, optimizer states, and activations for all parameters. • **Risk of Catastrophic Forgetting:** May degrade performance on original tasks by overwriting general knowledge.

Parameter-Efficient Fine-Tuning (PEFT)

PEFT techniques enhance LLM fine-tuning efficiency by focusing on a small subset of parameters.

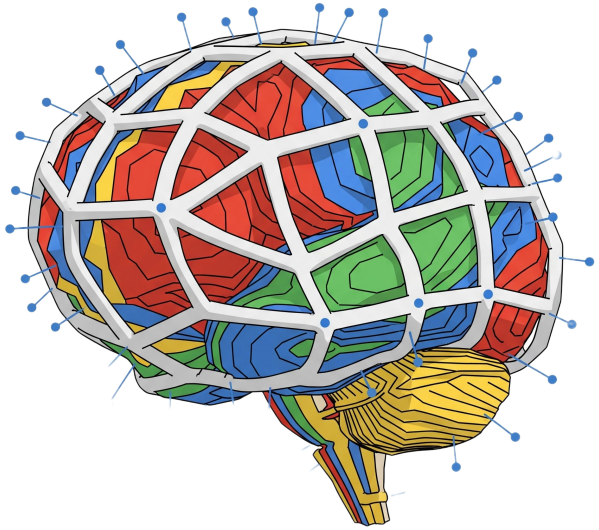
Pros	• **Reduced Cost:** Lower resource requirements means faster training, which also makes more experimentation feasible • **Portability:** Produces very small checkpoint files, which are easy to manage and distribute • **Mitigates Catastrophic Forgetting:** As most original weights are frozen
Cons	• **Potentially Lower Performance:** For tasks that are very different from the original pre-training data • **Risk of Overfitting:** PEFT's greater efficiency can sometimes lead to overfitting more quickly, requiring careful monitoring • **New Hyperparameters to Tune:** For example, the rank parameter used with LoRA techniques

advanced topic

A PEFT Technique: Low-Rank Adaptation (LoRA)

Complex high-dimensional updates to a model's weights can be approximated by using smaller, lower-rank matrices.

By specifying the **rank** of a LoRA adapter, you control the number of tunable parameters - how fine or coarse the LoRA "mesh" is.



Fine-tuning

Example: Full Fine-tuning

This guide walks you through how to fine-tune Gemma on a mobile game NPC dataset using **Hugging Face Transformers** and **TRL**.

As the model is small, this is a case where **full fine-tuning** is appropriate.

Full Model Fine-Tune using Hugging Face Transformers

goo.gle/gemma-ttt-finetuning-full

Fine-tuning

Example: PEFT

As earlier discussed, full fine-tuning of larger models is often impractical on a single GPU due to VRAM limitations.

PEFT techniques like LoRA **significantly reduce trainable parameters**, enabling fine-tuning on commodity hardware, which is crucial for this technical example.

Fine-tune Gemma in Keras using LoRA

goo.gle/gemma-ttt-finetuning-peft

Fine-tuning

Takeaways

- Fine-tuning is the process of of adapting a model to a specific task or domain
- Data for fine-tuning must be appropriately prepared
- Fine-tuning approaches vary based on model size and resources
  - **Full fine-tuning** is appropriate for **small models**
  - For **larger models**, **parameter-efficient approaches** become appropriate
  - For **even larger models**, **distributed approaches** can become appropriate
- Fine-tuning involves experimentation and evaluation to determine success

Learn more

Fine-tuning examples

- **Keras:** https://ai.google.dev/gemma/docs/lora_tuning
- **JAX:** https://gemma-llm.readthedocs.io/en/latest/colab_finetuning.html

Official Documentation

- Gemma @ AI Developers
  https://ai.google.dev/gemma/docs

Thank you!

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